AMRITA VISHWA VIDYAPEETHAM
DEEMED TO BE UNIVERSITY UNDER SECTION 3 OF UGC ACT, 1956
COIMBATORE CAMPUS

INSTITUTION'S INNOVATION COUNCIL
(Ministry of Education Initiative)

GENESIS

Genesis Ideathon: Final Presentation

Team ID: *GEN26-A0H-I143*

Team Name: *DEBUG AVENGERS*

Domain: *Digital Technologies, AI and Cybersecurity*

Amrita Vishwa Vidyapeetham, Coimbatore Campus

Date: 11th October 2026

Presentation Outline

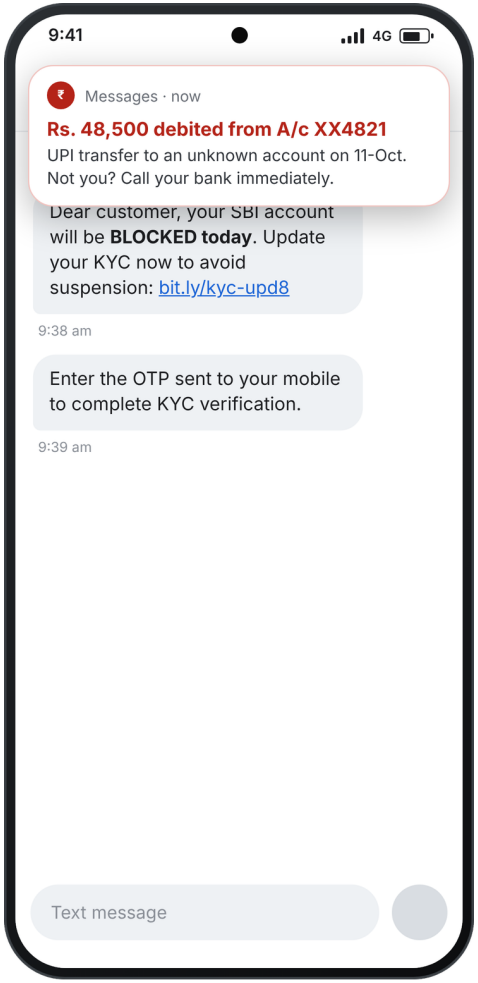
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Problem Statement

Making Digital Safety Understandable: AI-Powered Detection and Guidance for Phishing, Scams and Suspicious Digital Content

Every day, people in India get SMS and WhatsApp messages like *"Your SBI account will be blocked today, update KYC now"*. Some are real, many are scams, and most people can't tell the difference in the moment. They click, share an OTP or pay through UPI, and the money is gone before they realise it.

Today almost every protection we have works **after** the loss: you report on 1930 once the money has left your account. We want to solve the problem **before** it happens: warn people the moment a scam message arrives, in simple words and in their own language, and tell them what to do next.



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Problem Understanding

- **Who faces it?** Everyday smartphone users, mostly through SMS and WhatsApp. Students tend to be overconfident ("I'd never fall for this"), while elderly people trust anything that looks like it came from a bank or the police, and lose the most.

- **What causes it?** Scammers play on fake authority, urgency and greed, so people act in panic instead of thinking. Now they also use AI, so the messages have no spelling mistakes. In a 2025 test of 63 AI-written phishing emails, Outlook flagged only 4% as spam; about 96% reached the inbox. Many scams ("digital arrest", task jobs, UPI collect requests) don't even contain a link.

- **How does it affect people?** Indians lost about **Rs. 22,845 crore** to cyber fraud in 2024, roughly 3× more than in 2023. Cyber security incidents handled in India rose from 10.29 lakh (2022) to 22.68 lakh (2024). Beyond money, people lose trust, and many older users stop using digital payments.

Sources: MHA reply in Lok Sabha (July 2025); C. Opara et al., *Expert Syst. Appl.* 276, 127044 (2025); S. Zhou et al., *J. Inf. Technol. Case Appl. Res.* (2026); D. Debnath & S. Dutta, *Indian J. Law & Justice* 17(1).

Existing Gaps & Challenges

- **Existing tools each cover only one piece.** Google Messages scam detection works only inside Google's own SMS app. Airtel's AI blocks bad links, but only for Airtel users and only when there is a link. Blacklists lag behind new phishing links [1,2]. Chakshu and 1930 are vital for reporting, but people must first recognise the scam themselves.

- **Ground reality is different.** Indian scams come through SMS *and* WhatsApp, often in mixed Telugu, Hindi and English, and often with no link at all. Paste-and-check tools (like AuthentiScan Pro) only help if you already doubt the message, and people spot only about 40% of disguised links.

- **Main challenge still unsolved:** warning people *at the moment of risk*, across apps and languages, in a way they understand, without sending their private messages to a server and without crying wolf on genuine messages; reviews flag fast-changing attacks as open problems [2,3].

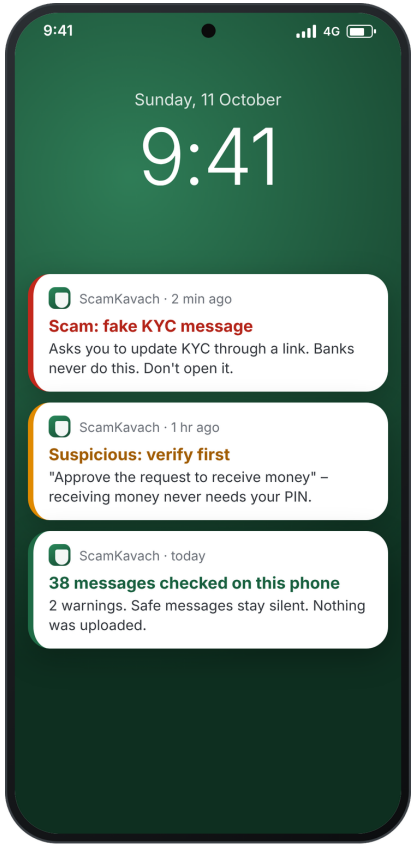
[1] A.K. Jain & B.B. Gupta, *Telecommunication Systems* 68, 687–700 (2018). [2] A. Safi & S. Singh, *J. King Saud Univ. Comp. & Info. Sci.* 35(2), 590–611 (2023). [3] Y.A. Kustiawan & K.I. Ghauth, *IEEE Access* 13 (2025). Also: Google (2024); MediaNama (2025); Costa (2026); Reynolds et al. (2020).

Our Proposed Solution

- **ScamKavach: an on-device scam shield built into the phone.** Installed once and with the user's permission, it runs in the background and checks incoming SMS and WhatsApp notifications. No need to copy-paste anything.

- **How it works:** each message is checked on the phone in under a second. **Safe** → stays silent. **Suspicious** → a small "verify first" banner. **Scam** → a warning card with the risk level, 2–3 simple reasons, the trick being used, and one safe step (e.g. *"Banks never ask for OTP. Don't click. Report on 1930."*). It can read the warning aloud in Telugu or Hindi.

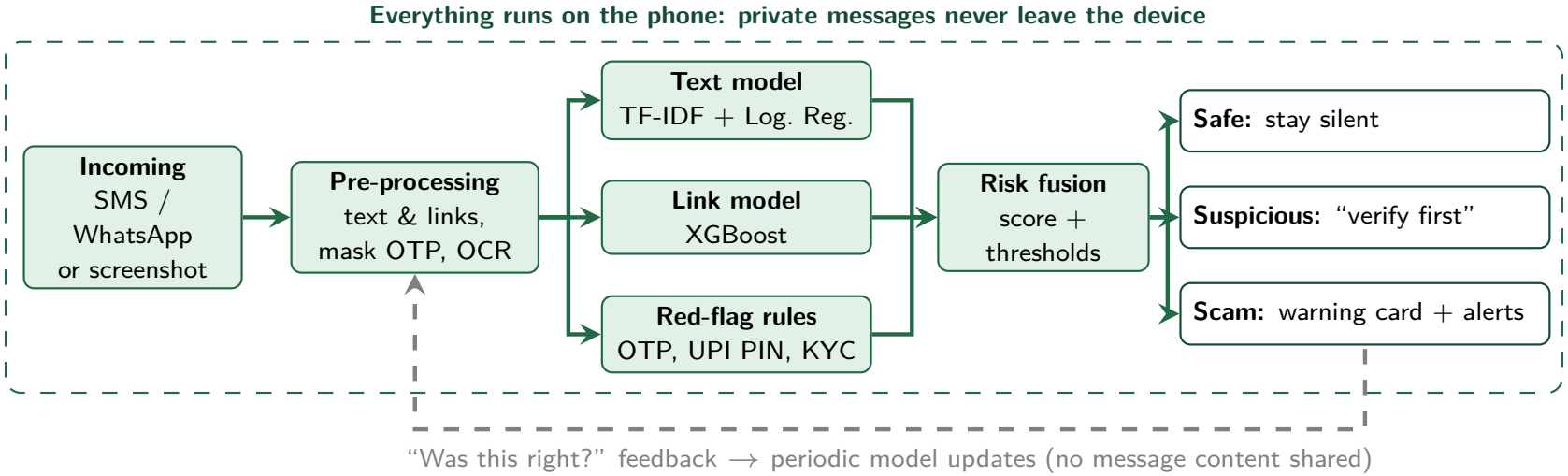
- **Value to users:** protection *before* the loss, not after. Planned safety nets: a pause screen before opening a UPI app after a flagged message, and an optional family alert for elderly users. Every warning names the trick, so people slowly learn to spot scams on their own.



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ScamKavach **complements** existing protections like Google Messages, Airtel and 1930. It fills the gaps between them, without claiming to catch every scam.

Technical Approach



Proven lightweight models (no new algorithm) + our own Indian scam rules and language data. The link model reads only the URL text and never opens the link. Tested on unseen messages with precision, recall, F1-score and false-positive rate, not accuracy alone.

Tech Stack

Frontend: Android (Kotlin), Text-to-Speech

Backend: Kotlin background service; FastAPI for updates

Database: Room (SQLite), stored on the phone

AI/ML, APIs, or Hardware: scikit-learn, XGBoost, ONNX Runtime Mobile, ML Kit OCR

Innovation & Uniqueness

- **What is new?** One on-device shield that works *across* SMS and WhatsApp on any Android phone and any network, understands code-mixed Telugu/Hindi/English, and catches scams even when there is *no link*, which link blockers miss completely.

- **Why this approach fits:** the research is clear that people rarely check messages themselves, so the warning must come to them at the moment of risk. Running small models on the phone keeps it fast, works offline, and keeps private messages private. Warnings follow the format studies found works best: risk level + 2–3 reasons + one next step.

- **Key differentiator:** ScamKavach doesn't just block, it **explains the trick** ("fake authority", "urgency", "greed"). With the planned UPI pause and family alert, it protects people at the exact moment money is about to move, and teaches them something with every warning.

Sources: E. Bouma-Sims et al., IEEE S&P (2026); P. Buono et al., CHI EA (2023); S. Zhou et al. (2026); A.N. Osijirin et al., *Saudi J. Eng. Technol.* 11(4), 355–361 (2026).

Feasibility & Scalability

- **Feasibility:** Everything we need is free or already with us: laptops, open-source tools and a few Android phones. *Prototype:* label 300+ real messages (Telugu, Hindi, English), train the models, and build notification-based checking plus a "share to check" option for links and screenshots, using Android's Notification Listener with user consent. *Planned next:* UPI pause, family alert and voice warnings, after consent design and testing with 10 users.

- **Scalability:** Models are a few MB and run on the phone, so new users add almost no server cost. New languages and scam types need only more data. Later, the engine can be offered as a plug-in for phone makers, bank and UPI apps.

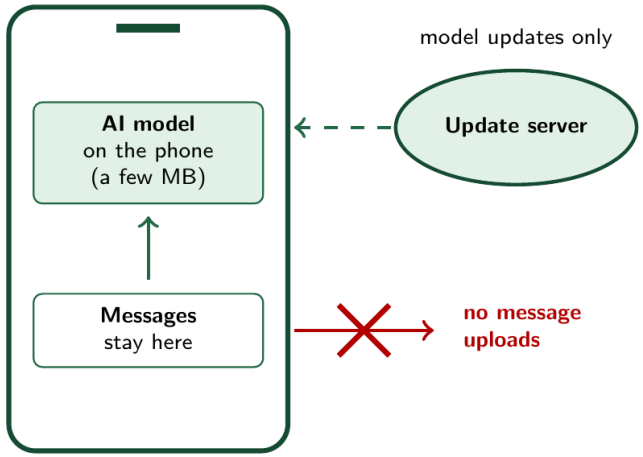
- **Risks & mitigations:** *Privacy fears* $\rightarrow$ on-device checking, masking, clear consent (in line with India's DPDP Act 2023). *False alarms* $\rightarrow$ three levels, tuned thresholds, never delete a message. *Unsafe links* $\rightarrow$ links are never opened and page scripts never run. *Scammers adapt* $\rightarrow$ user feedback and regular model updates. *iPhone limits* $\rightarrow$ Android first.

Cost for Product (Optional)

- Development or hardware cost: about **Rs. 18,000**: two budget Android test phones (≈Rs. 8,000 each) and the one-time Google Play developer fee (US$25, ≈Rs. 2,200). Laptops and all software tools are already available or open-source.
- Hosting or maintenance cost: about **Rs. 800 per month** (≈Rs. 10,000 a year) for a small cloud server that only sends model updates. Since all checking happens on the phone, there is no per-message AI or API cost.
- Estimated total: about **Rs. 28,000 for the first year** (INR). The prototype itself can be built at near-zero cost using free cloud tiers.

Estimates as of October 2026; phone and server prices are indicative.

Why it stays cheap: the checking runs on the user's phone, so the cost does not grow with every message checked.

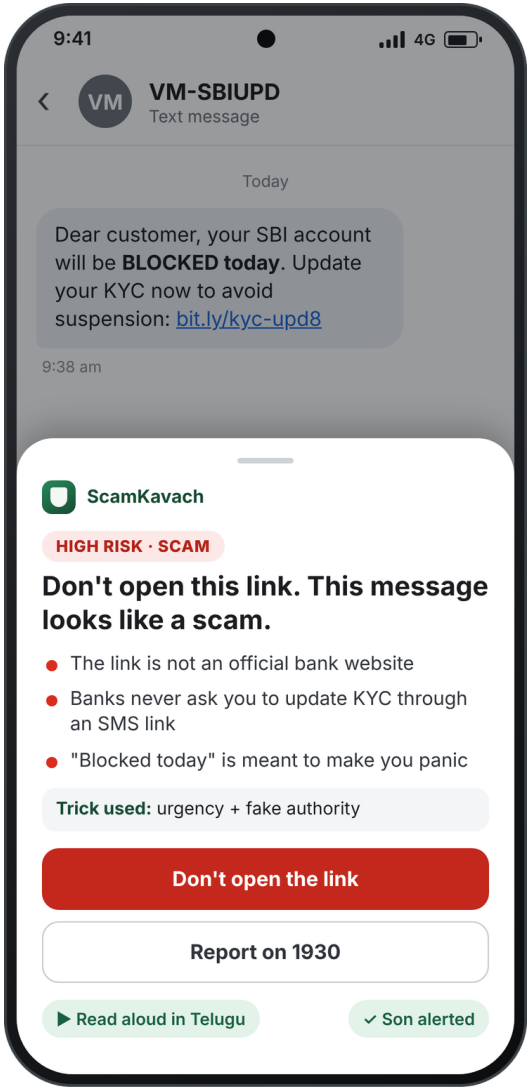


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Use Case 1: Fake KYC SMS

An elderly parent

- **User:** Lakshmi, a 62-year-old retired teacher in Vijayawada. She is new to UPI and reads Telugu more easily than English.
- **Scenario:** An SMS says her SBI account will be blocked today unless she updates KYC through a short link.
- **What ScamKavach does:** the link model sees a lookalike, non-official website, and the rules catch “KYC” and “blocked today”. It shows a **Scam** card, reads it aloud in Telugu, and alerts her son.
- **Outcome:** she doesn’t click, and her son calls her within minutes. Her savings stay safe.

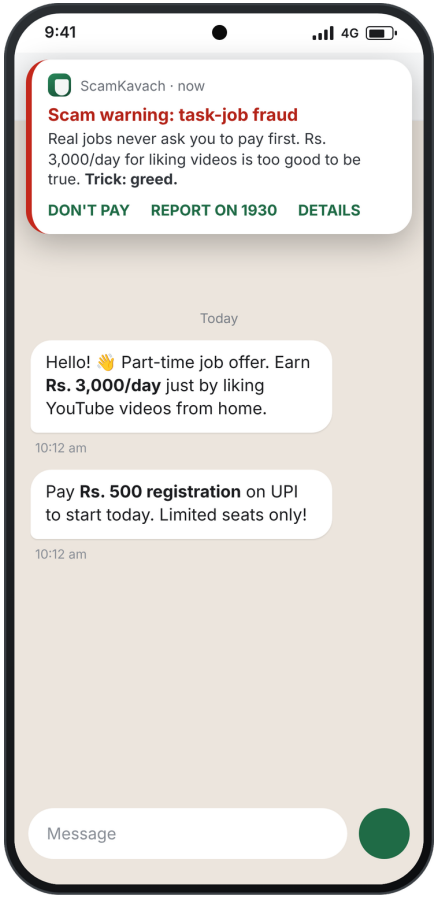


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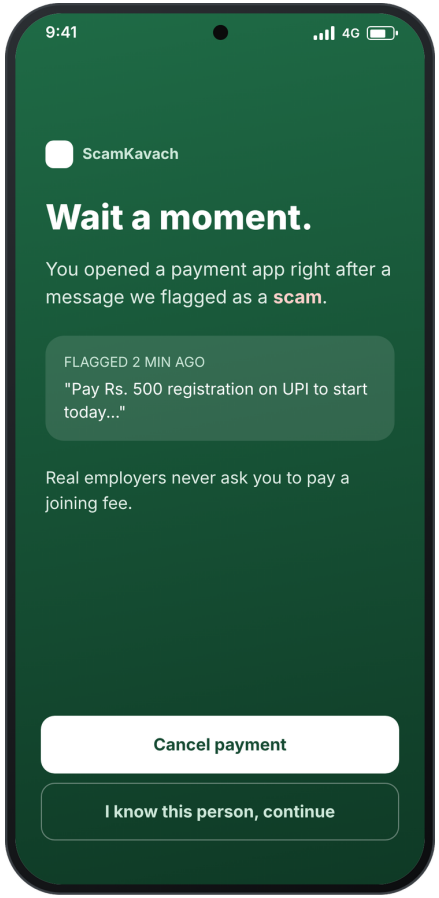
Use Case 2: Task-Job Scam

A college student

- **User:** Arjun, a first-year engineering student looking for pocket money. He is confident he can't be fooled.
- **Scenario:** A WhatsApp message offers Rs. 3,000 a day for liking videos, but asks for a Rs. 500 "registration fee" on UPI. There is **no link**, so link blockers miss it.
- **What ScamKavach does:** the text model and red-flag rules mark it **Scam** and name the greed trick. When Arjun opens his UPI app a few minutes later, a pause screen asks him to stop and think.
- **Outcome:** he doesn't pay, and he now knows what a task scam looks like.



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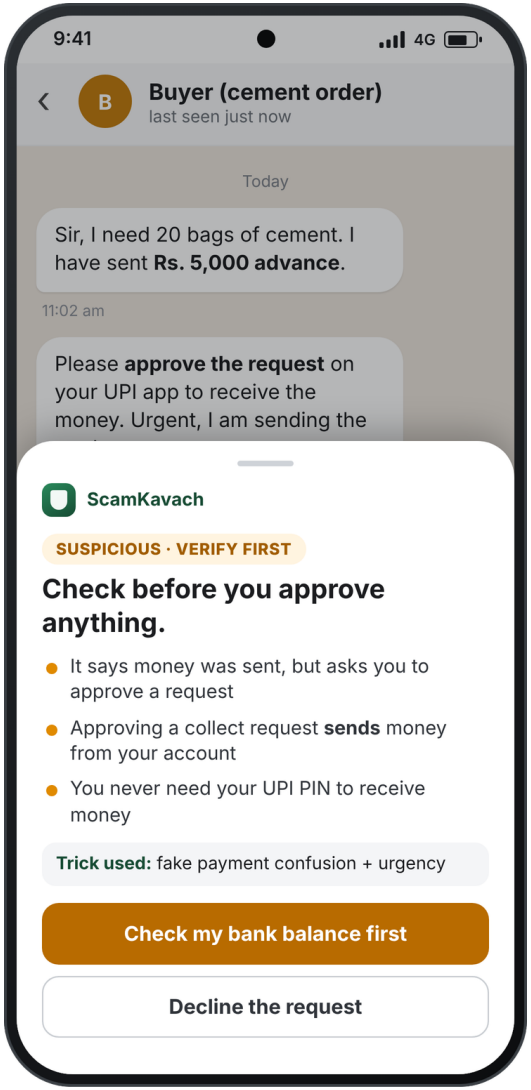


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Use Case 3: Fake UPI Payment

A small shop owner

- **User:** Ramesh, who runs a steel and cement shop and takes many UPI payments every day.
- **Scenario:** A "buyer" says he sent a Rs. 5,000 advance for cement and asks Ramesh to approve the request on UPI. It is really a **collect request** that would *take* money from him.
- **What ScamKavach does:** rules catch "approve the request" together with "sent money" and show a softer **Suspicious** card.
- **Outcome:** he checks his balance, declines the request, and keeps his Rs. 5,000.



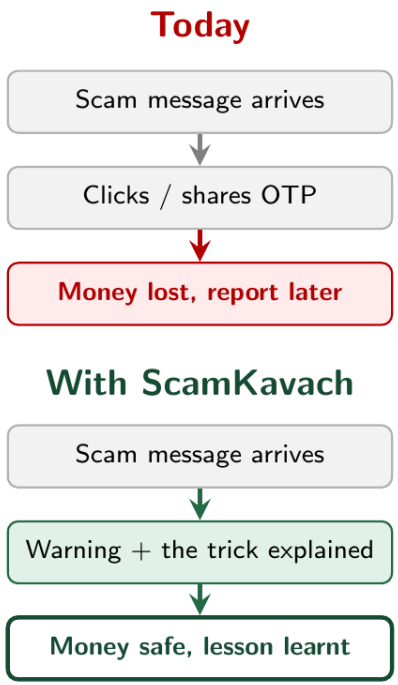
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Impacts and Benefits

- **For users:** people are warned before they click, pay or share an OTP, in their own language and without needing to be tech-savvy. Elderly users get a family safety net, and every warning teaches them a little more about how scams work.

- **Social and economic impact:** fewer families losing savings, fewer complaints for the 1930 helpline and cyber cells, and more trust in UPI and digital payments, especially among first-time and elderly users who are most likely to give up on them after a scam.

- **Measurable outcomes (prototype targets):** on 300+ unseen real messages, recall of at least **90%** (scams caught) and a false-positive rate under **5%** (genuine messages wrongly flagged), with precision and F1-score reported too; at least **8 out of 10** testers, including elderly users, understand the warning without any help; check each message in under **1 second** on a budget phone.



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Sustainable Development Goals (SDGs)

SDG Number	Target Number	Goal Name	Connection
SDG 16	Target 16.4	Peace, Justice and Strong Institutions	Helps fight organised scam networks by stopping fraud before money is stolen.
SDG 8	Target 8.10	Decent Work and Economic Growth	Builds trust in UPI and digital banking, so more people stay in the formal financial system.
SDG 4	Target 4.4	Quality Education	Every warning explains the trick, building real digital safety skills among youth and adults.

Individual Member Contribution

Member	Contribution or responsibility
Ketharaju Bhargav	Team lead: problem framing, literature survey of 13 papers and reports, presentation.
Kelam Rahul	User flow: warning levels, warning card, UPI pause, family alert, user testing.
Lalam Tejash	Technical approach: tool comparison, model choice, on-device Android plan.
Guggilla Raketh Sastry	Ground research: Indian scam samples, statistics, scam patterns, data labelling.

Yukti ID

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